

Problem 8

Suppose that exactly one of three devices being tested is defective, and let this be event A . Event B means that a device is functional. What do events $A + B$ and AB represent?

Solution

We need to carefully interpret events A and B , then determine what $A + B$ and AB represent.

Step 1: Clarify the Events

Event A

The problem states: "exactly one of three devices being tested is defective, and let this be event A ."

This is somewhat ambiguous. Let me interpret it as:

Interpretation: Event A = "exactly one of the three devices is defective"

This means:

- Device 1 is defective and devices 2, 3 are functional, OR
- Device 2 is defective and devices 1, 3 are functional, OR
- Device 3 is defective and devices 1, 2 are functional

Event B

Event B = "a device is functional"

This typically means: "the device being tested (or selected) is functional"

However, this is also ambiguous. Let me consider two interpretations:

Interpretation 1: B = "a specific device (e.g., the one currently being tested) is functional"

Interpretation 2: B = "at least one device is functional"

Given the context, Interpretation 1 seems more natural.

Step 2: More Precise Formulation

Let me reformulate more carefully:

Sample Space: We have three devices. Each device can be either functional (F) or defective (D).

Event A : Exactly one device is defective (and two are functional).

Event B : When we test a randomly selected device, it is functional.

Step 3: Analyze Event $A + B$ (Union)

Definition

$$A + B = A \cup B$$

This represents the event that "A occurs OR B occurs (or both)."

When Does $A + B$ Occur?

The event $A + B$ occurs when:

- Exactly one device is defective (event A), OR
- The selected device is functional (event B), OR
- Both

Analysis

If event A occurs (exactly one device is defective), then:

- Two devices are functional
- One device is defective
- If we randomly select a device, the probability it's functional is $\frac{2}{3}$

Note that if A occurs, B may or may not occur (depending on which device is selected).

Interpretation of $A + B$

$A + B =$ "Either exactly one device is defective, or the selected device is functional (or both)"

More simply: $A + B$ occurs in all cases except when:

- Either zero devices are defective or two/three devices are defective (not A), AND
- The selected device is defective (not B)

Step 4: Analyze Event AB (Intersection)

Definition

$$AB = A \cap B$$

This represents the event that " A occurs AND B occurs."

When Does AB Occur?

The event AB occurs when:

- Exactly one device is defective (event A), AND
- The selected device is functional (event B)

Analysis

If event A occurs, there are three possible scenarios:

1. Device 1 is defective, devices 2 and 3 are functional
2. Device 2 is defective, devices 1 and 3 are functional
3. Device 3 is defective, devices 1 and 2 are functional

For AB to occur, A must occur AND the device we select must be functional.

Interpretation of AB

$AB =$ "Exactly one device is defective AND the selected device is functional"

This means:

- Among the three devices, exactly one is defective
- When we test/select a device, we happen to pick one of the two functional ones

Step 5: Relationship Between Events

Is $B \subset A$?

No. Event B (selected device is functional) can occur even when A doesn't occur (e.g., all three devices are functional).

Is $A \subset B$?

No. Event A (exactly one defective) can occur even when B doesn't occur (e.g., we happen to select the defective device).

Are A and B Independent?

No. The occurrence of A affects the probability of B :

- If A occurs: $P(B|A) = \frac{2}{3}$ (two functional out of three)
- If all devices are functional: $P(B|\text{all functional}) = 1$
- If all devices are defective: $P(B|\text{all defective}) = 0$

Step 6: Example

Suppose the three devices are in states: (F, F, D) where exactly one is defective.

- Event A occurs (exactly one defective)
- If we select device 1 or 2: Event B occurs, so AB occurs
- If we select device 3: Event B does not occur, so AB does not occur

Event $A + B$ occurs in all three cases (because A always occurs).

Final Answer

Event $A + B$:

$A + B = \text{"Exactly one device is defective OR the selected device is functional (or both)"}$

This event occurs unless:

- The number of defective devices is not exactly one (i.e., 0, 2, or 3 devices are defective), AND
- The selected device is defective

Event AB :

$AB = \text{"Exactly one device is defective AND the selected device is functional"}$

This event occurs when:

- Among the three devices, exactly one is defective and two are functional
- When we test or select a device, we happen to select one of the two functional devices

In Probabilistic Terms: If devices are randomly distributed and one is randomly selected:

- $P(A + B) = P(A) + P(B) - P(AB)$ (inclusion-exclusion)
- $P(AB) = P(A) \cdot P(B|A) = P(A) \cdot \frac{2}{3}$